Azerbaijani property listings

From-scratch decision trees and Pegasos SVMs

Price regression and premium-tier classification

MLE-AI-201 Machine Learning. Team project. 17 October 2026

Review draft: confirm full author names and effort split before submission.

One dataset, two prediction tasks

Kaggle bina.az sale listings; asking prices in AZN

Dataset stage	Rows / definition
Raw file	100,775 listings
After URL deduplication	64,454 listings
Train / validation / test	45,118 / 9,667 / 9,669
Premium threshold	210,000 AZN (training median)

Regression predicts sale price. Classification predicts above-median price.

Preparation and fair comparison

A single seeded split supports all headline results

14 encoded features from listing attributes

Training-only medians, category vocabulary and SVM scaling

Validation selects settings; test rows are evaluated once

Three-fold CV refits preprocessing inside each training fold

Leakage guard: remove unit_price and total_price, which derive from the target.

Decision tree built in NumPy

Shared split search supports both targets

Sort feature values and test midpoints between distinct values

Classification uses Gini or entropy; regression uses squared error

Stop by depth, leaf size or impurity improvement

Validation chose depth 12 for regression and 10 for classification

The from-scratch trees nearly match the corresponding sklearn baselines.

Pegasos SVM built in NumPy

Linear hinge-loss classifier plus nonlinear RFF extension

Labels: standard = -1, premium = +1

Stochastic subgradient updates with decreasing step size

RFF approximates an RBF kernel before Pegasos fitting

Validation selected lambda, update budget and RFF bandwidth

RFF improves test F1 from 0.777 to 0.806 on the integrated split.

Held-out price regression

Test: 9,669 listings; three-fold CV uses training rows only

Model	MAE (k AZN)	RMSE (k AZN)	R^2	CV log-RMSE
Our tree	103.3	759.1	0.227	0.409 ± 0.018
sklearn tree	103.5	759.2	0.227	0.408 ± 0.018

Close baseline agreement; the luxury tail dominates RMSE.

Errors are back-transformed and reported in AZN.

Held-out premium classification

Premium means price above the 210,000 AZN training median

Model	Test F1	Test AUC	3-fold CV F1
Our tree	0.904	0.967	0.897 ± 0.001
sklearn tree	0.904	0.967	0.897 ± 0.001
Our linear Pegasos	0.777	0.826	0.774 ± 0.013
sklearn SGD hinge	0.778	0.839	0.778 ± 0.000
Our RFF Pegasos	0.806	0.874	0.807 ± 0.006

The tree leads on F1. RFF adds 2.9 percentage points over linear Pegasos.

Where the models fail

Luxury listing prices produce disproportionate regression error

85 test listings above the 2.5M AZN training 99th percentile

3.64M AZN mean absolute error in that tail

72.0k AZN mean absolute error for the other 9,584

90.6% of high-price listings are underpredicted

Next: verify suspicious listings and test on a later market scrape.

Team work and reproducibility

Responsibilities visible in Git history; percentages need team agreement

GitHub handle	Main contribution
nhesen	Data cleaning, features and EDA
BulqeyisMammadova	Decision tree, experiments and forest bonus
Ulvi022	Pegasos SVM, RFF and SVM experiments
daduu16	Integration, metrics, baselines, report and results

Reproduce with python -m src.run_all. All 19 integration tests pass.